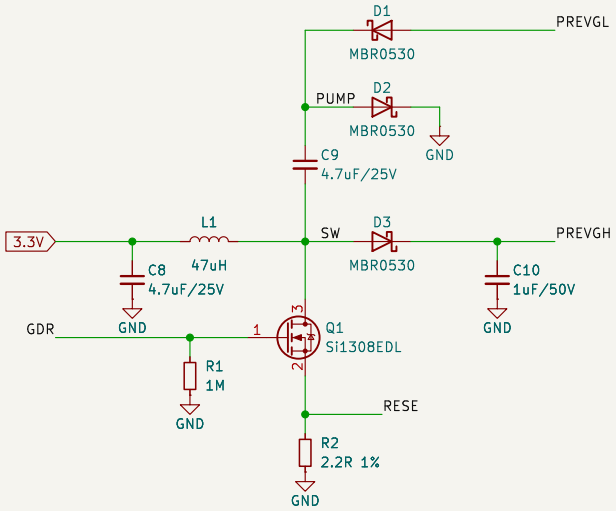
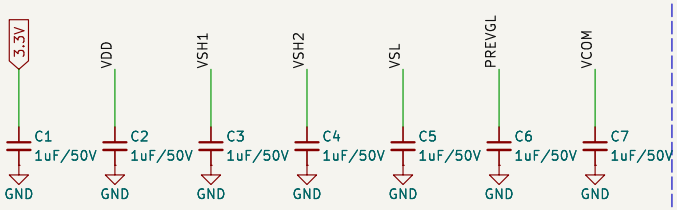
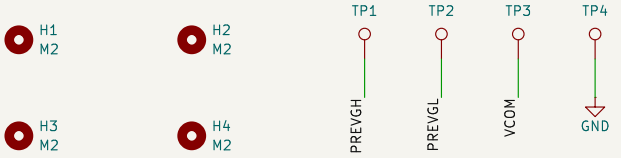
Boost: VGH/VGL generation (GDEY0213B74 reference circuit)



Panel decoupling (X5R/X7R, >=25V, close to EP_FPC1)

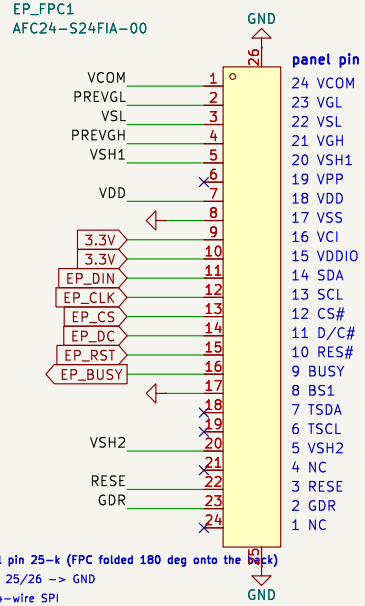


Mounting & test points



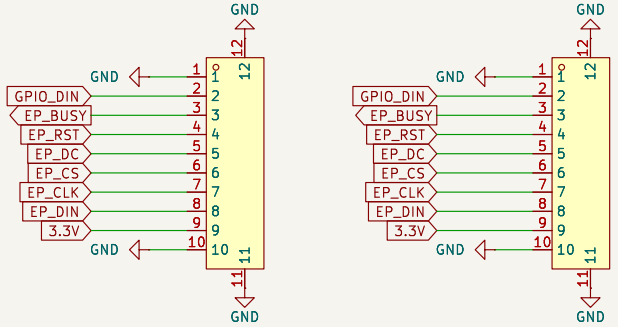
H1–H4: M2 screws + standoffs onto the 18650 holder. Back-side parts must stay below the standoff height (EP_FPC1 = 2.0mm).

EP_FPC1: panel FPC 24P 0.5mm (back side)



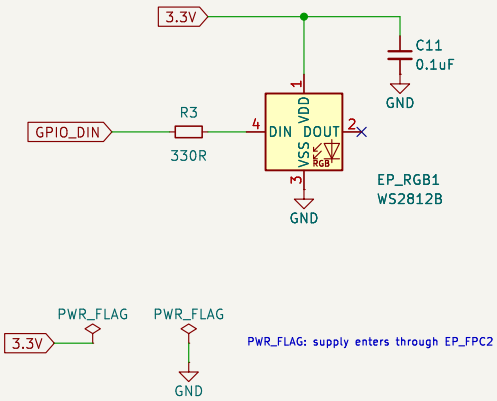
Pad k = panel pin 25−k (FPC folded 180 deg onto the back)
Mounting pads 25/26 -> GND
BS1 = L -> 4-wire SPI

EP_FPC2 (right) / EP_FPC3 (left): to keyboard, 10P 0.5mm, in parallel



Same pinout as 1.02inch-right EP_FPC2. Populate both (or only the side used):
EP_FPC2 = right half, EP_FPC3 = left half (mirror position on the PCB)

EP_RGB1: optional status LED (reverse-mount 6028, as on 1.02inch)



PWR_FLAG: supply enters through EP_FPC2

NOTES
1. GDEY0213B74: SSD1680, 250x122, VCI 2.2–3.7V (3.3V here). BS1 = GND -> 4-wire SPI, so EP_DC is required.
2. Boost/charge pump = Good Display reference circuit: L1 47uH (Isat >= 0.4A), Q1 Si1308EDL (alt. AO3400A, SOT-23), D1–D3 MBR0530, R1 1M gate pull-down, R2 2.2 ohm RESE sense. (2.2 ohm + 47uH suit SSD ICs; UC ICs use 0.47 ohm + 10uH.)
3. Caps on PREVGH/PREVGL/VSH/VSL/VCOM must be rated >= 25V (1uF/50V X5R; C8/C9 4.7uF/25V X5R).
4. Pins 1/4 (NC), 6/7 (TSCL/TSDA), 19 (VPP) are left open.
5. Stack-up: panel taped to the FRONT, every part on the BACK. The panel FPC wraps 180 deg around the board edge into EP_FPC1, so its gold fingers face AWAY from the PCB: EP_FPC1 must be top- or dual-contact. AFC24-S24FIA-00 is dual-contact, front-insert / rear-flip: the FPC enters on the side opposite the pads, so face that side to the edge. The bottom-contact AFC01 of the 1.02inch board won't mate. Because of the fold, connector pad k = panel pin 25-k.
6. EP_FPC2/EP_FPC3 pinout = 1.02inch-right EP_FPC2. They sit at mirrored positions (x = +/-6 mm), same orientation, so one board fits a left and a right half; the keyboard socket is mirrored.
7. Keep the L1/Q1/R2/C8 loop tight; route PREVGH/PREVGL/VCOM short and away from EP_CLK/EP_DIN; solid GND on the back.
8. R3 (WS2812 data series R) may be 0R; EP_RGB1 is optional.

Keyboard link: EP_FPC2 10P 0.5mm FPC, same pinout as 1.02inch-right
Panel on front (double-sided tape), all parts on back, M2 standoffs above 18650

ErgoCai

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Title: 2.13inch e-Paper adapter – GDEY0213B74

Size: A4	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 1/1